



QP CODE: 24806044



Reg No :

Name :

M.C.A. DEGREE EXAMINATION, NOVEMBER 2024

First Semester

Faculty of Technology & Applied Science

Master of Computer Application

CORE - MCACT102 - DIGITAL LOGIC & COMPUTER ORGANIZATION

2020 Admission Onwards

B2FF8B3C

Time: 3 Hours

Maximum: 75 Marks

Part A

*Answer any **ten** questions*

*Each question carries **3** marks*

1. Explain parity row and parity column in block parity.
2. Explain the use of hamming code.
3. What is ASCII code?
4. What is a Multiplexer?
5. Discuss the working of SR flipflop.
6. What is a Universal shift register?
7. Define clock rate.
8. Briefly discuss bit-ORing.
9. What are interrupts?
10. What are the different types of buses used in DMA?
11. Define parallel processing.
12. Define instruction level parallelism.

(10×3=30 marks)





Part B

Answer *all* questions

Each question carries **9** marks

13. a) Perform subtraction using 2's complement method on the binary and explain the steps.

i) $1011011011 - 101101$

ii) $10101011 - 101010$

iii) $11.10 - 110.1$

OR

- b) Detect and correct errors in the following even parity hamming code 0101101

14. a) Solve the following using K-Map. $F(A,B,C, D) = \sum(1,3,9,11,4, 5, 12, 13, 10,14)$.

OR

- b) Write notes on half adder and 4-input multiplexer.

15. a) Describe the various addressing modes.

OR

- b) Discuss in detail about the hardwired control unit with block diagram

16. a) Explain Random Access Memory in detail.

OR

- b) Explain the working of Interrupts.

17. a) Explain in detail the following SIMD parallel structures: Array Processors, Vector Processors

OR

- b) Explain the interconnection structures used in multiprocessors .

(5×9=45 marks)

